

ASSIGNMENT – 17

ADVANCED PROGRAMMING

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SESSION-SPRING

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```

assignment17.py > RegistrationService > register_user
1  import re
2
3
4  class InvalidEmailError(ValueError):
5      def __init__(self, email):
6          super().__init__(f"Invalid email address: {email}")
7
8
9  class UnderageError(Exception):
10     def __init__(self, age):
11         super().__init__(f"User age {age} is below the minimum required age of 18")
12
13
14 class RegistrationService:
15
16     def register_user(self, email: str, age: int) -> bool:
17
18         # Internal assertion
19         assert isinstance(age, int), "Age must be an integer"
20
21         # Empty email check
22         if email is None or email.strip() == "":
23             raise InvalidEmailError(email)
24
25         # Regex email validation
26         pattern = r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'
27
28         if not re.match(pattern, email):
29             raise InvalidEmailError(email)
30
31         # Age validation
32         if age < 18:
33             raise UnderageError(age)
34
35         return True
36

```

```

assignment17.py > RegistrationService > register_user
35     return True
36
37
38 # User Input Section
39 service = RegistrationService()
40
41 try:
42     email = input("Enter Email: ")
43     age = int(input("Enter Age: "))
44
45     result = service.register_user(email, age)
46
47     if result:
48         print("Registration Successful")
49
50 except Exception as e:
51     print("Error:", e)

```

TESTING USING PYTEST-

```
test_assignment17.py > test_invalid_email_none
1  import pytest
2  from ass17 import (
3      RegistrationService,
4      InvalidEmailError,
5      UnderageError
6  )
7
8
9  @pytest.fixture
10 def service():
11     return RegistrationService()
12
13
14 def test_successful_registration(service):
15     assert service.register_user("user@example.com", 20) is True
16
17
18 def test_invalid_email_empty(service):
19     with pytest.raises(InvalidEmailError):
20         service.register_user("", 22)
21
22
23 def test_invalid_email_format(service):
24     with pytest.raises(InvalidEmailError):
25         service.register_user("wrongemail.com", 25)
26
27
28 def test_underage_user(service):
29     with pytest.raises(UnderageError):
30         service.register_user("user@example.com", 16)
31
32
33 def test_invalid_email_none(service):
34     with pytest.raises(InvalidEmailError):
35         service.register_user(None, 19]
```

OUTPUT-

```
C:\Users\VITU\OneDrive\Desktop\c>python -u "c:\Users\VITU\OneDrive\Desktop\c\assignment17.py"
Enter Email: Vitu@gmail.com
Enter Age: 19
Registration Successful

C:\Users\VITU\OneDrive\Desktop\c>python assignment17.py
Enter Email: bhitali
Enter Age: 7
Error: Invalid email address: bhitali

C:\Users\VITU\OneDrive\Desktop\c>
```